EE115B *Digital Circuits*

Instructor: Chenxi Xiao

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Hengzhao Yang
Zhifeng Zhu
Yajun Ha

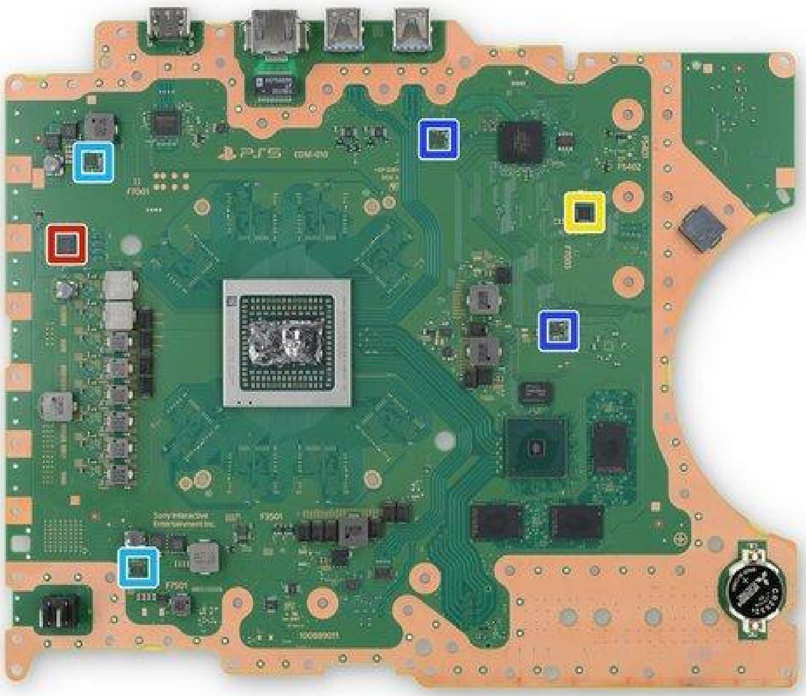
上海科技大学
ShanghaiTech University

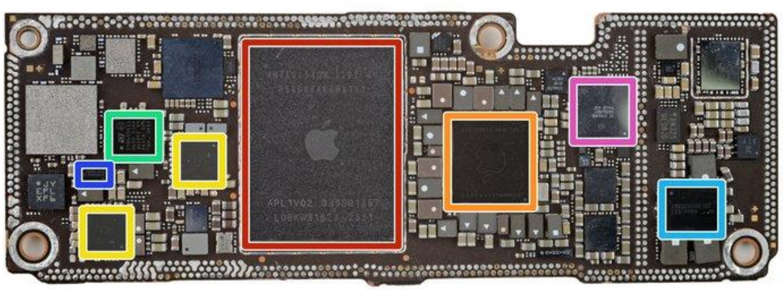
Part 1

Course Information

Why digital circuits?

Digital chips are so commonly used everywhere.





https://www.ifixit.com/Teardown/PlayStation+5+Teardown/138280

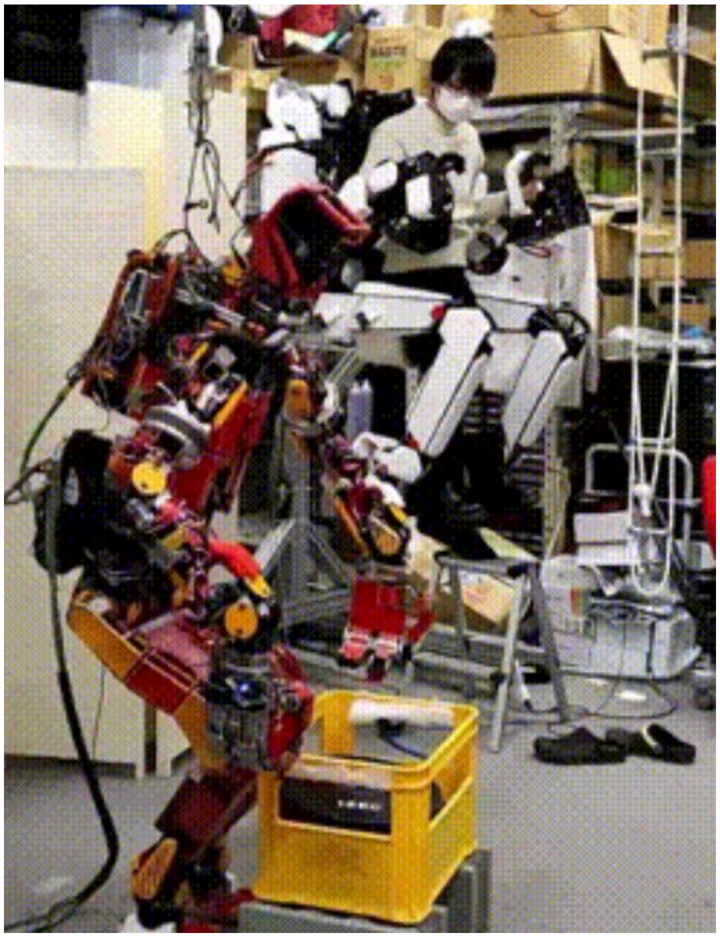
https://www.ifixit.com/Guide/iPhone+15+Pro+Max+Chip+ID/165320

Why digital circuits?

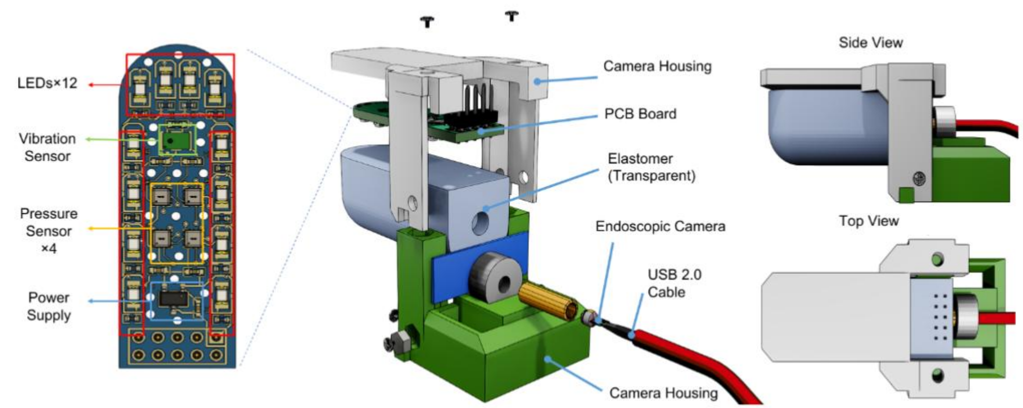
For computational purposes: Processors, RAM, Flash, ...

Many Systems are Hybrid

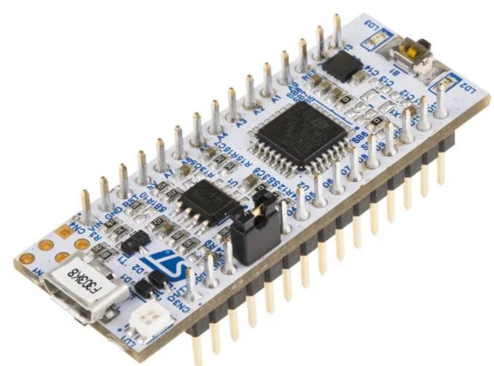
Robots



Sensors



MCU with ADC/DAC



Earphones

https://arxiv.org/abs/2410.10353
https://haraduka.github.io/jaxon-tablis-imitation/

https://sg.rs-online.com/web/p/microcontroller-development-tools/9092862

The position of this course

A **fundamental**, **EE** course

- EE111 Electric Circuits
- EE115A Analog Circuits

  => EE112 Analog Integrated Circuits
- EE115B Digital Circuits

  => EE113 Digital Integrated Circuits

  => Embedded Systems

  => FPGA ...

  => Board level circuit design

Course Information

Course Instructor: Chenxi Xiao 肖晨曦

Office: SIST 1D-301A

email: xiaochx@shanghaitech.edu.cn

My Course Table   Print

Academic Year & Semester: 2025-2026Year&Term2 Go

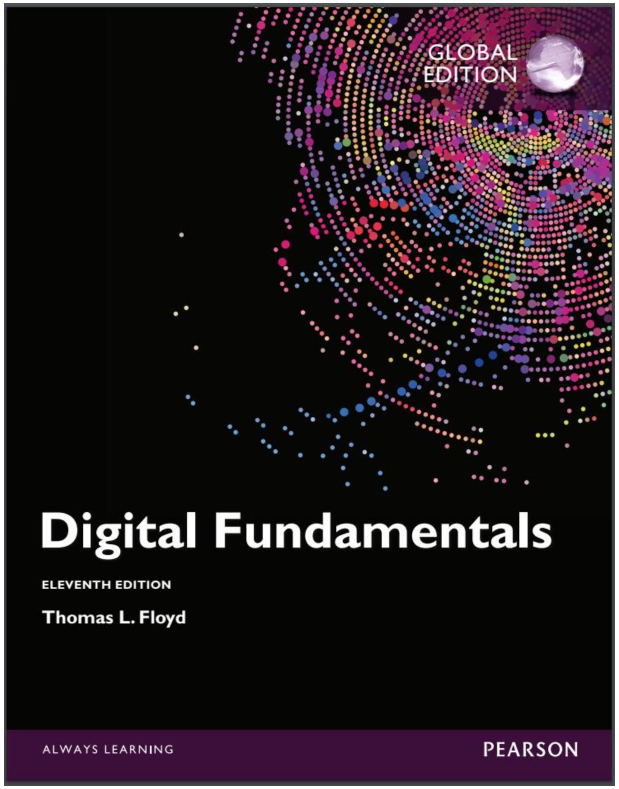
Units/Weeks	Mon	Tue	Wed	Thur	Fri	Sat	Sun
1 8:15 - 9:00					Digital Circuits(EE115B.01) (1-15,信息学院 1D104)		
2 9:10 - 9:55							
3 10:15 - 11:00		Digital Circuits(EE115B.01) (Odd1-15,信息学院 1D104)					
4 11:10 - 11:55							

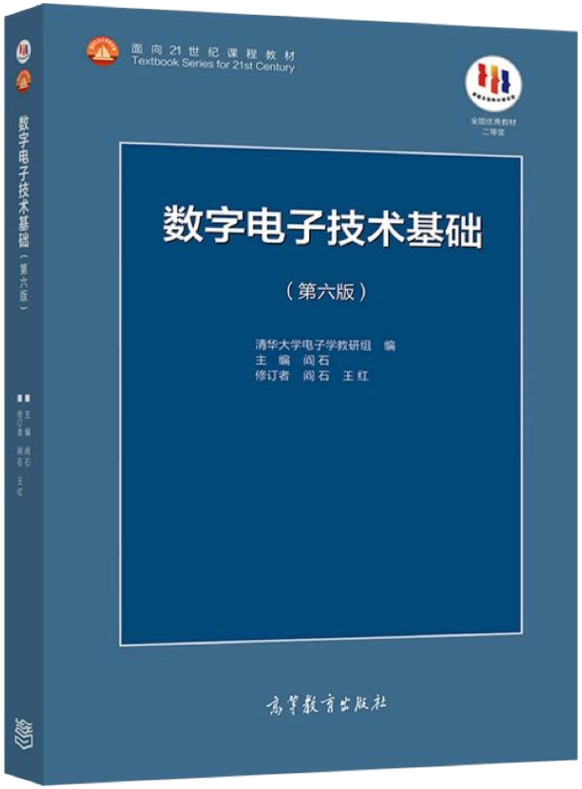
Experimental Course Instructor: Juan Li 李娟

Office: 1B-204

email: lijuan1@shanghaitech.edu.cn

Textbooks





Thomas L. Floyd,
Digital Fundamentals

阎石
数字电路技术基础

Syllabus

Chapter Name	Main Teaching Content (Main Knowledge Points)
1	Numbering, Coding, Logic Gates
2	Boolean Algebra and Logic Simplification
3	Logic Conversion
4	Combinational Logic Circuits: Adders, Comparators, Code Converters, Multiplexers, etc.
5	VHDL Hardware Description Language Overview
6	Sequential Logic Circuits: Latches, Flip-Flops, Counters, Registers, State Machines
7	Data Storage Circuits
8	Signal Processing Circuits

Grading Scheme:

- Attendance: 5%
- Homework: 25%
- Laboratory: 15%
- Midterm Exam: 25%
- Final Exam: 30%

Syllabus

Grade:

- Letter grade is based on ranking.

Integrity

- Do not copy other people's homework.
- Responsible use of tools (Including AI)

Office Hours

Office Hour: Friday 9:55 am - 10:25 am
Appointment based, Immediately after class

TAs: appointment based. Send email first:

Zhengying Zhu 朱政颖 (Exams)
zhuzhy12025@shanghaitech.edu.cn

Ruilin Zhang 张瑞麟 (HW1-2)
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Yongzhi Zhang 张雍治 (Projects)
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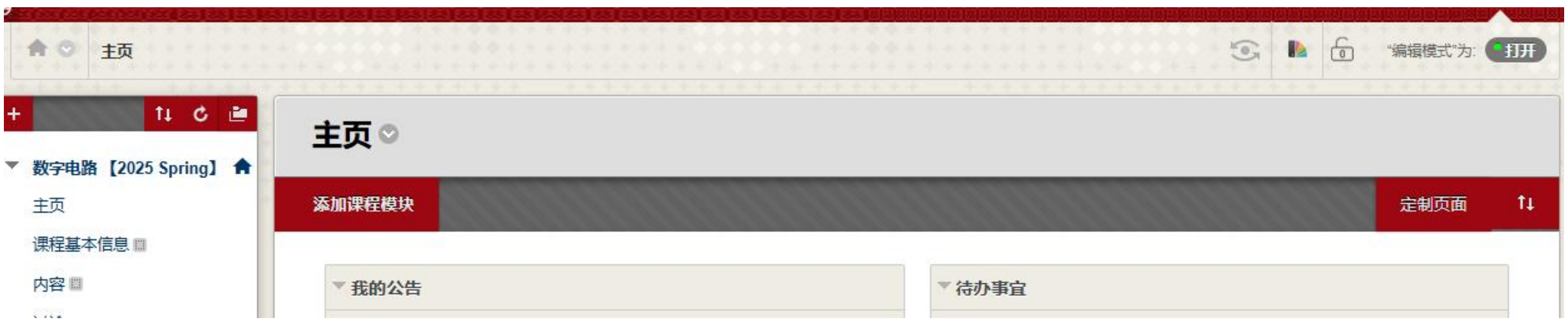
Tiantian Yang 杨天添 (HW3-4)
yangtt2023@shanghaitech.edu.cn

Resources

Blackboard:

- Official annocements
- Publish course slides
- Homework submission



Feishu Group

- General, unoffical QA
- No important annocements
- (for grade-related questions, please inquiry via email, by Week 17)

Join Feishu QA group

Chapter 1: Number Systems and Coding

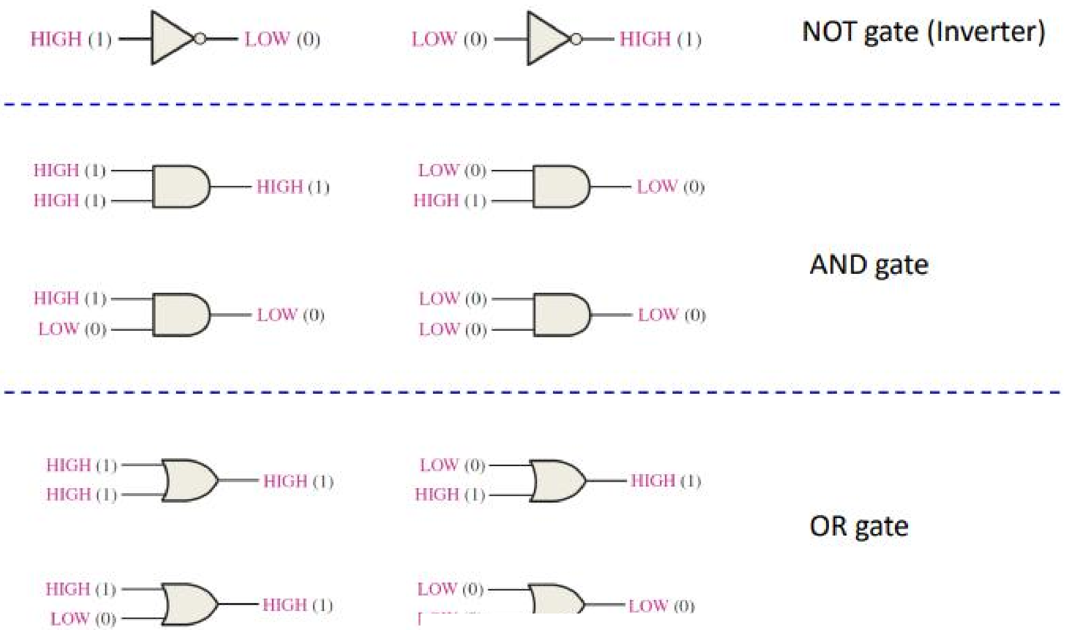
Digital quantity, binary, octal, hex, ...

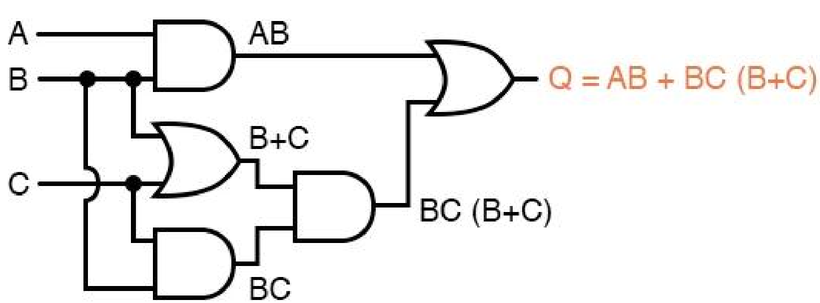
Decimal	Binary	Octal	Hex
0	0000	0	0
1	0001	1	1
2	0010	2	2
3	0011	3	3
4	0100	4	4
5	0101	5	5
6	0110	6	6
7	0111	7	7
8	1000	10	8
9	1001	11	9
10	1010	12	A
11	1011	13	B
12	1100	14	C
13	1101	15	D
14	1110	16	E
15	1111	17	F

Chapter 2-3: Boolean Algebra, Logic Circuit Simplification

1. basic logic
2. logical expressions

& simplification





AB + BC (B + C)

↓ Distributing terms

AB + BBC + BCC

↓ Applying identity AA = A to 2nd and 3rd terms

AB + BC + BC

↓ Applying identity A + A = A to 2nd and 3rd terms

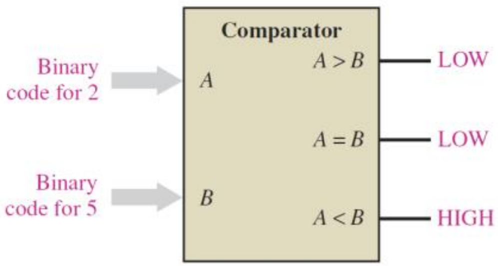
AB + BC

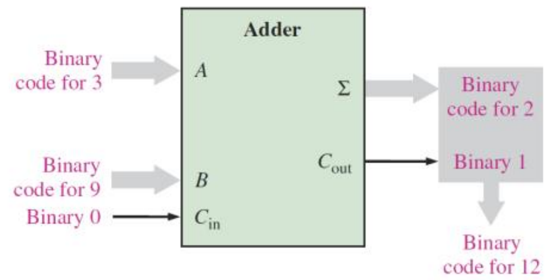
↓ Factoring B out of terms

B (A + C)

Chapter 4: Combinational Logic

- The output only dependent on the present input

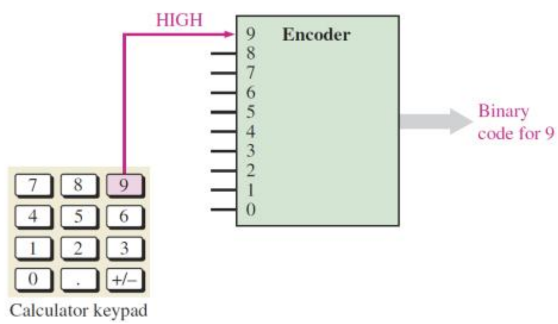


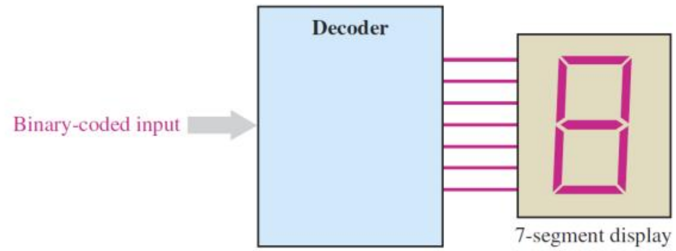


Subtraction

Multiplication

Division...

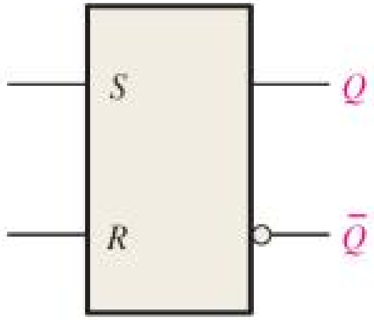


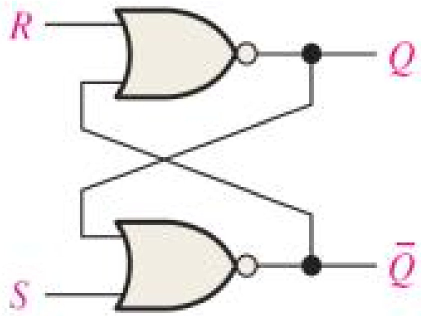


Chapter 6: Sequential Logic

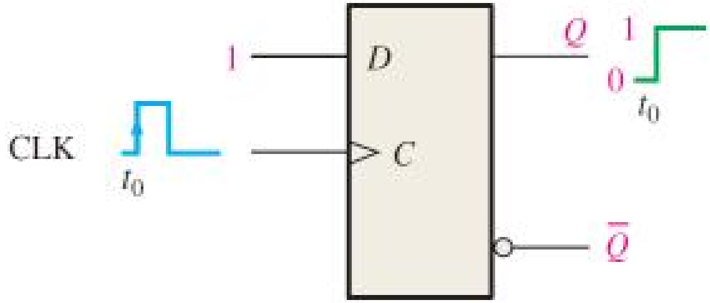
- The output depends not only on the present input but also on the history of the input

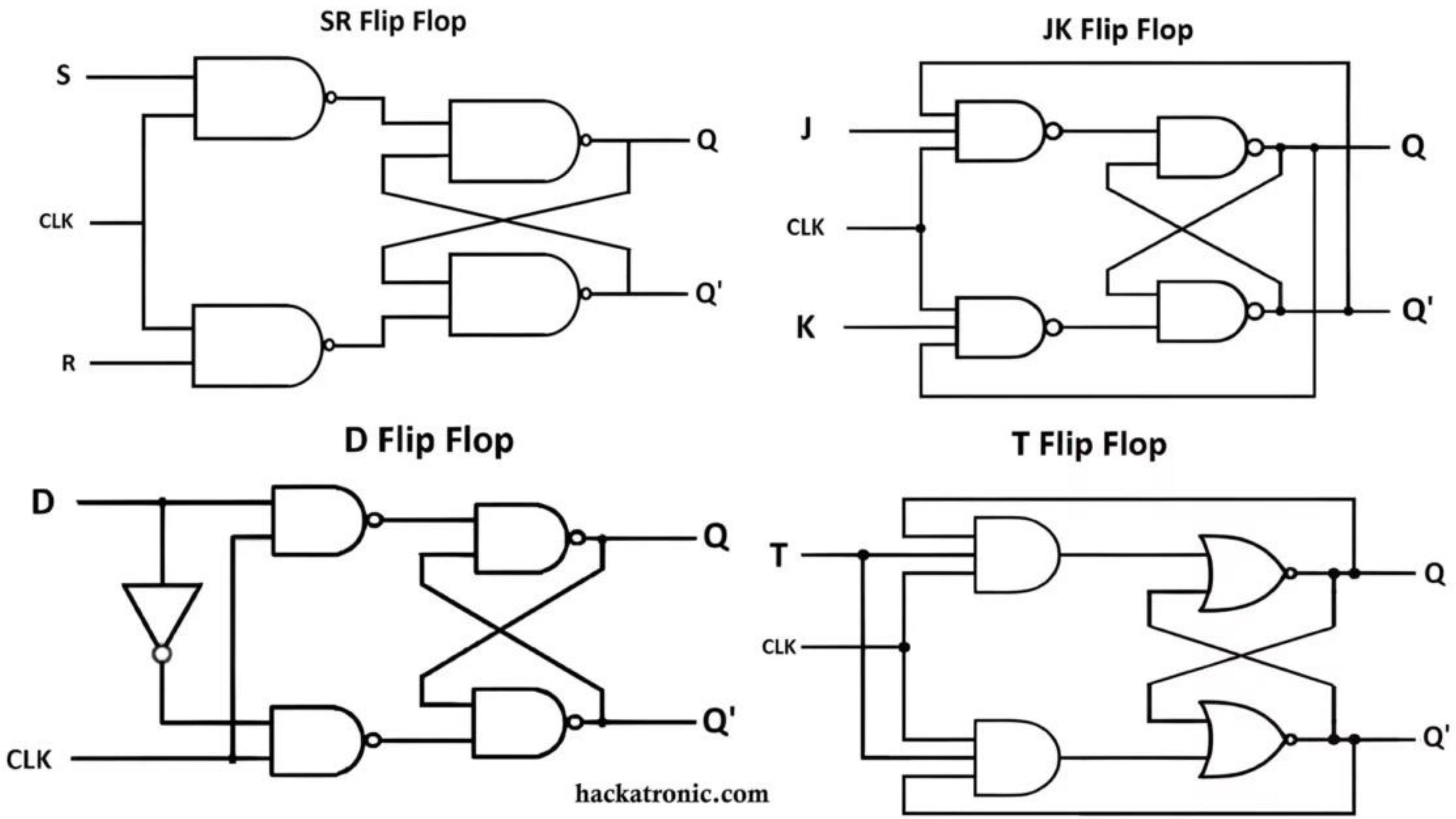
Latch





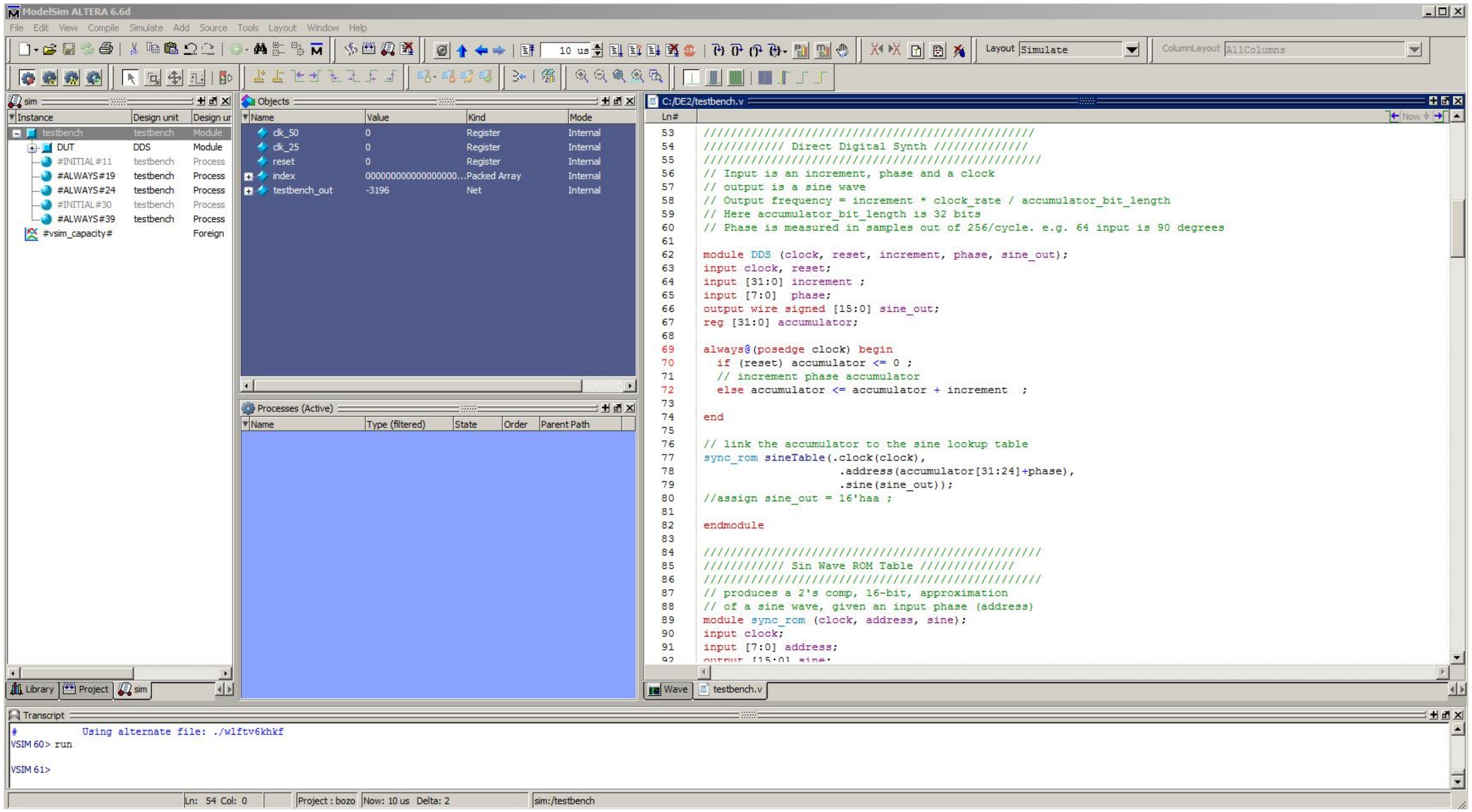
Flip-Flop





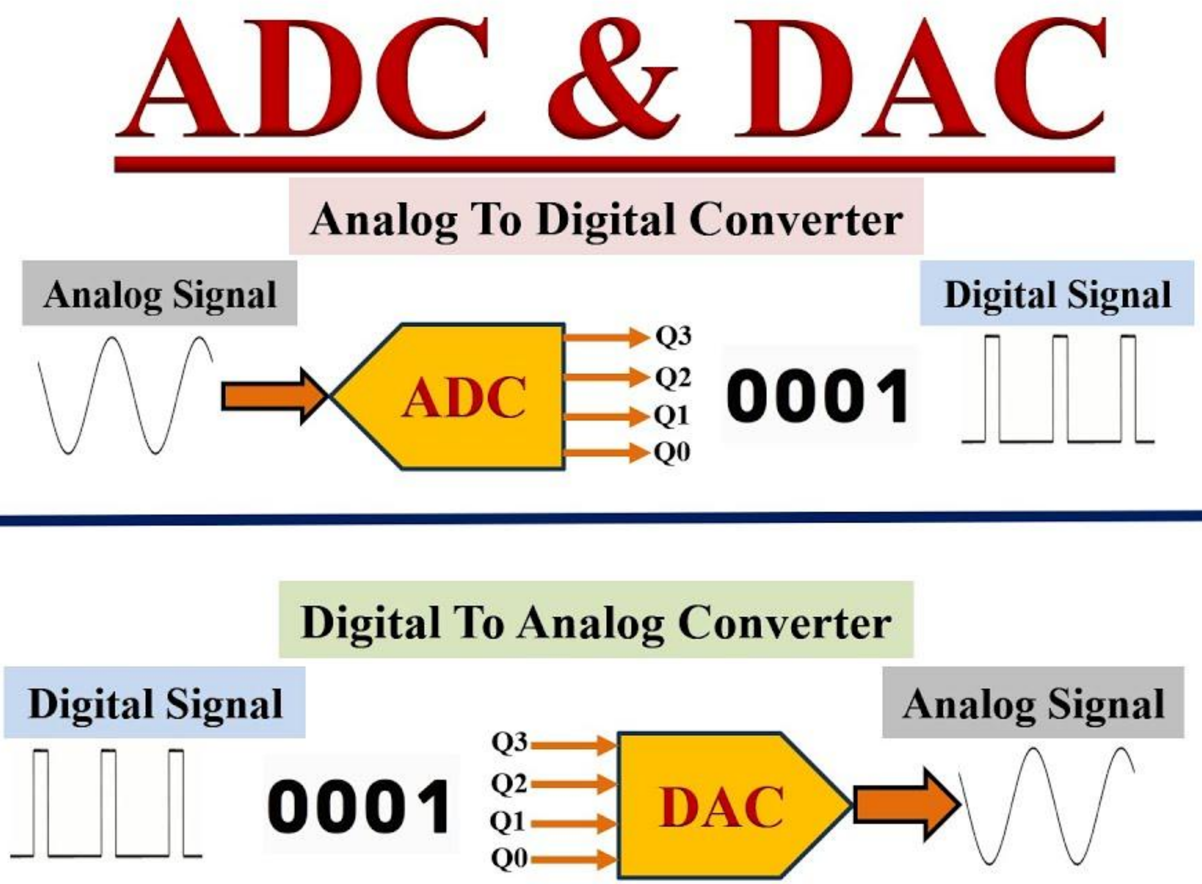
https://www.hackatronic.com/what-is-flip-flop-circuit-truth-table-and-various-types-of-flip-flops/

Chapter 5 Hardware Description Language



https://people.ece.cornell.edu/land/courses/ece5760/ModelSim/index.html

Chapter 8: Signal Conversion



https://www.google.com/url?sa=i&url=https%3A%2F%2Fwww.youtube.com%2Fwatch%3Fv%3DplLDN3mFY24&psig=AOvVaw367j_h1UzvOyeXy6sXYSYc&ust=1739451180063000&source=images&cd=vfe&opi=89978449&ved=0CBQQjRxqFwoTCJiyhOqWvosDFQAAAAAdAAAAABAE